# Aerothermodynamic Propulsion, Active Ground-Effect Suction, 1,048-Part Parametric CAD Architecture, and Autonomous VLA Control of the 2026 Next-Generation Tesla Roadster

Chief Engineer Technical Monograph: Cold-Gas Thruster Supersonic Nozzle Mechanics, 10,000-PSI COPV Blowdown Thermodynamics, 2G Inverted Suction Physics, Multi-Modal LLM/VLA Flight-Track Autonomy, and 10-Environment Proving Ground Validation

Following the live prototype flight and inverted-wall demonstration at the SpaceX Rocket Development and Test Facility in McGregor, Texas on October 15, 2026, the second-generation Tesla Roadster has redefined the physical boundaries of terrestrial and near-surface vehicular locomotion. By coupling a 1-megawatt rear-wheel-drive electric powertrain with four 20-kilowatt underbody ground-effect extraction fans and ten SpaceX-engineered cold-gas supersonic thrusters fed by 600 liters of high-pressure nitrogen, the platform decouples longitudinal acceleration and lateral cornering from classical Coulomb tire friction constraints (CellCog Engineering Analysis; EVXL Technical Report).

| Engineering Domain | Primary System Specification | Verified Performance Envelope | Primary Source Attribution |
|---|---|---|---|
| **Tiered Trim Architecture** | Performance (2-seat), Thrust Augment (2-seat), Rocket Augment (1-seat) | 0–60 mph: <1.60 s (Performance), <0.90 s / 0.85 s (Thrust), <0.59 s / 0.55 s (Rocket) | Tesla Roadster Portal; Kingy AI |
| **Electric Traction Drive** | Dual rear motors (no mechanical differential), 8.4:1 planetary gearsets, cobalt-steel rotors | >1.0 MW (1,341–1,500 hp), >1,400 Nm torque, 800–1,000 V bus, 500 kW V4 Supercharging | EVXL; Hagerty / Top Gear Interviews |
| **Cold-Gas Thruster (CGT) Array** | 4 main rear diffuser De Laval nozzles + 6 perimeter lift/yaw nozzles (Optimus electric valves) | >80 kN (18,000 lbf) net thrust, supersonic Mach diamond plume, 54–70.7 s sea-level $I_{sp}$ | Tech Insider; Paul Tan |
| **High-Pressure COPV Storage** | 200 L trunk COPV (Thrust) + 400 L cabin COPV replacing passenger seat (Rocket = 600 L total) | 10,000 psi (690 bar) structural rating; 5,500 psi (379 bar) Track readout; ~227–250 kg $N_2$/air | CellCog; EVXL |
| **Ground-Effect Suction System** | 4 × 20 kW rear diffuser fans + 4 deployable rubber/nylon-bristle active sealing skirts | 4,700 kgf (46.1 kN / 2.0g) stationary downforce; 170° inverted wall climb at 21 m height | Drive Tesla Canada; US Patent 12,377,920 |
| **Active Aero & Chassis Mass** | Optimus ankle-actuated 700 kg rear wing (90° airbrake), 80° 5-surface front wing, barge boards | <1,700 kg dry target mass (~2,350 kg wet flight mass); Ti-6Al-4V, Mg-AZ91D, T1100G carbon | Teslarati; EVXL |

| Engineering Domain | Primary System Specification | Verified Performance Envelope | Primary Source Attribution |
|---|---|---|---|
| **3D CAD & Autonomy Stack** | 1,048-part parametric CAD assembly + multi-modal LLM/VLA planner + 1 kHz SpaceX GNC loop | Autonomous McGregor flight demo; 7 m convoy kickflip; 3.3g retro-landing at 6.5 m | CBC News; Electrek |

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## 1. Executive Technical Synthesis and the October 2026 McGregor Proving Ground Demonstration

On the evening of October 15, 2026, at the 4,000-acre SpaceX Rocket Development and Test Facility in McGregor, Texas, Tesla and SpaceX jointly demonstrated the redesigned second-generation Tesla Roadster in a live autonomous proving ground sequence that fundamentally departed from conventional automotive engineering (Paul Tan Automotive News; Teslarati). Preceded by a static fire of a SpaceX Raptor methalox rocket engine, the vehicle executed four distinct physical maneuvers under closed-loop computer control: an inverted vertical wall climb to a pitch angle of 170 degrees at an altitude of 21 meters while sustaining 4,700 kilograms-force (46.1 kN) of underbody fan suction; a zero-speed inverted release followed by a 3.3g cold-gas retro-thruster arrest at 6.5 meters above ground level; a standing launch to 60 mph (96.56 km/h) in 0.81 seconds inside a confined launch pad; and a 7-meter-high airborne 360-degree axial roll ("kickflip") over a stationary convoy comprising a Tesla Semi, Cybertruck, Cybercab, Model Y, and Model 3 (CBC News; Electrek).

For over a century, terrestrial vehicle dynamics have been governed by a single immutable boundary condition: the Coulomb friction limit of pneumatic elastomers against asphalt. As Tesla Vice President of Vehicle Engineering Lars Moravy noted during post-reveal technical briefings, even ultra-high-performance track tires such as the Pirelli P Zero Trofeo RS "Elect" plateau at a peak longitudinal and lateral friction coefficient of approximately $\mu \approx 1.26$ on unprepped road surfaces (CellCog; Paul Tan). Because normal load on a unbanked road without aerodynamic assistance equals vehicle weight $W = m\ g$, maximum tire-limited acceleration $a_{max} = \mu\ g$ is invariant to motor torque or vehicle mass once wheel-slip saturation occurs. This physical ceiling explains why quad-motor electric hypercars exceeding 1,400 kW—such as the Rimac Nevera R (1.66 s with rollout) and Xiaomi SU7 Ultra (1.98 s)—have asymptotically converged near 1.65 to 1.90 seconds for the 0–60 mph sprint (Kingy AI).

The 2026 Tesla Roadster architecture circumvents this asymptotic friction ceiling through two orthogonal physical mechanisms developed collaboratively between Tesla Vehicle Engineering and SpaceX Propulsion (Tesla; Tech Insider). First, an active underbody boundary-layer extraction system—comprising four 20 kW electric axial-centrifugal fans and four independent compliant rubber and nylon-bristle sealing skirts—evacuates air beneath the floor pan to maintain a sub-ambient pressure differential of 10.2 to 13.2 kPa (EVXL; CellCog). This artificial vacuum multiplies the normal force $F_N$ on the contact patches by up to 3.0x at zero forward velocity (1.0g gravity plus 2.0g suction), tripling the available tire tractive force before wheel spin initiates. Second, the vehicle integrates a SpaceX-derived cold-gas reaction and main propulsion system utilizing Composite Overwrapped Pressure Vessels

(COPVs) storing up to 600 liters of dry nitrogen or compressed atmospheric air at working pressures of 5,500 psi (379 bar) and structural vessel ratings up to 10,000 psi (690 bar) (Kingy AI; EVXL). Expelling this dense gas through supersonic converging-diverging De Laval nozzles generates over 80 kN (18,000 lbf) of direct momentum thrust—four to five times the tractive force of the driven rear tires—completely independent of road contact (Teslarati; Tech Insider).

## 2. Tiered Vehicle Architecture: Performance, Thrust, and Rocket Variants

Rather than offering a single monolithic vehicle specification, Tesla structured the October 2026 Roadster program into a base road-legal platform (Performance) and two progressive aerospace hardware augments (Thrust and Rocket), with series production scheduled for 2028 at Giga Texas (Tesla; Kingy AI). While an entry-level 800 kW (1,073 hp, 885 lb-ft) base development mule was referenced in engineering briefings with Top Gear and InsideEVs, the three publicly announced tiers establish a clear division between road-homologated ground-effect traction, track-focused longitudinal rocket augmentation, and full 6-degree-of-freedom (6-DOF) airborne trajectory control (EVXL).

| Subsystem / Parameter | Performance (Base Tier) | Thrust Augment (Race Ready) | Rocket Augment (Air Ready) |
|---|---|---|---|
| **Cabin & Seating Layout** | 2-Seat Cockpit (Driver + Passenger) + Rear Luggage Bay (fits 2 rollaboards) | 2-Seat Cockpit (Driver + Passenger); Trunk occupied by 200-L COPV | 1-Seat Single-Seater; Passenger bay occupied by secondary 400-L COPV |
| **Official 0–60 mph Claim** | < 1.60 seconds (avg. 1.71g; 21.46 m rollout distance) | < 0.90 seconds official / 0.85 s Hagerty / 0.81 s McGregor live demo (3.38g) | < 0.59 seconds official / 0.55 s rollout interview (avg. 4.64g; 7.91 m) |
| **Top Speed Envelope** | 210–214 mph (338–344 km/h) aero-limited baseline | > 230 mph (370 km/h) certified; 330–383 mph (Mach 0.5) engineering target | > 230 mph (370 km/h) official; 300–330 mph pressure-assisted regime |
| **Cold-Gas Thruster Array** | None (Pure electric RWD + Active Aero + Underbody Suction) | 4 Rear Diffuser Main Supersonic Cold-Gas Thrusters (>80 kN total) | 10 Perimeter Cold-Gas Thrusters (4 Rear Main + 4 Wheel-Well Lift + 2 Nose Yaw) |
| **COPV Propellant Volume** | 0 Liters | 200 Liters (53 US Gal) carbon-wrapped tank in rear trunk | 600 Liters Total (200 L trunk + 400 L cabin tank; ~227–250 kg $N_2$/air) |
| **Underbody Suction System** | 4 × 20 kW extraction fans + 4 deployable rubber/nylon-bristle skirts | 4 × 20 kW extraction fans + 4 deployable rubber/nylon-bristle skirts | 4 × 20 kW extraction fans + 4 deployable rubber/nylon-bristle skirts |
| **Electric Traction Drive** | Dual rear motors, >1.0 MW (1,341–1,500 hp), >1,400 Nm, 8.4:1 planetary | Dual rear motors, >1.0 MW (1,341–1,500 hp), >1,400 Nm, 8.4:1 planetary | Dual rear motors, >1.0 MW (1,341–1,500 hp), >1,400 Nm, 8.4:1 planetary |
| **Curb Mass (Dry / Loaded)** | ~1,680 kg dry / ~1,700 kg ready | ~1,810 kg dry / ~1,900 kg with 83 kg $N_2$ charge | ~2,080 kg dry / ~2,330–2,350 kg with 250 kg $N_2$ charge |
| **Reservation & Homologation** | $50,000 ($5k down + $45k in 10 days); Road Ready homologation | $250,000 ($5k down + $245k in 10 days); Limited Release (1 per buyer) | Invite Only; Explicitly Marked "Not Approved for Public Road Use" |

The structural divergence across the three variants illustrates a deliberate trade-off between grand-touring utility and impulse density (CellCog). Whereas the 2017 Roadster concept promised a 2+2 seating configuration and a 200 kWh battery pack yielding 620 miles (998 km) of range, the 2026 engineering program eliminated the rear seats and reduced nominal driving range to 310–320 miles (499–515 km) in order to achieve a dry vehicle mass just under 1,700 kg (EVXL; Paul Tan). In the Rocket configuration, the passenger seat and right-hand footwell controls are removed entirely to accommodate a 400-liter cylindrical-spherical composite pressure vessel alongside the 200-liter rear trunk vessel, transforming the cabin into a single-pilot aerospace cockpit (Kingy AI; EVXL).

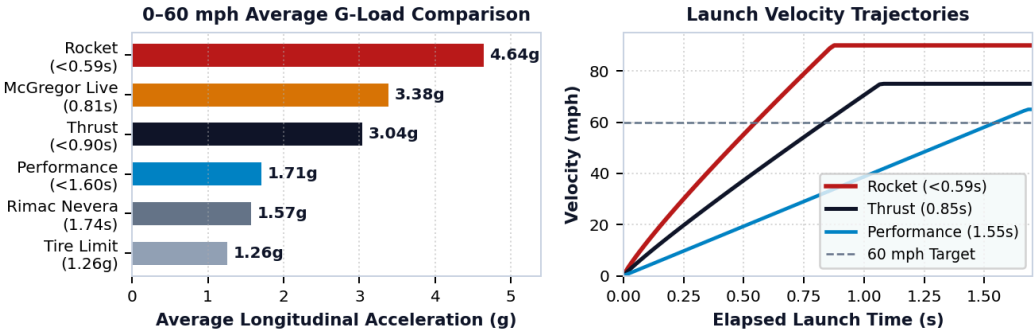



Figure 1: Comparative longitudinal acceleration (in g) and velocity-time profiles across tire-limited hypercars and the three 2026 Tesla Roadster tiers (Performance, Thrust, and Rocket), grounded in October 2026 reveal telemetry (CellCog; Kingy AI).

## 3. High-Voltage 4680 Powertrain, Dual-Motor Torque Vectoring, and Thermal Architecture

Contrary to the tri-motor and quad-motor all-wheel-drive architectures prevalent in contemporary electric hypercars, the 2026 Roadster deletes the front drive unit completely to free frontal packaging volume for the five-element active front wing, radiator exhaust ducting, nose yaw thrusters, and movable pedal-box assembly (EVXL). All electrical traction power is delivered to the rear axle via a compact twin-motor Rear Drive Unit (RDU) rated at over 1.0 megawatt (1,341 mechanical horsepower, with peak transient ratings reaching 1,100 kW / 1,500 hp) and over 1,400 Nm (1,033 lb-ft) of combined shaft torque (Kingy AI; EVXL). According to engineering display plaques at the McGregor unveil and interviews with MotorTrend and Hagerty, this dual-motor RDU delivers 33% to 50% more power than the Model S Plaid tri-motor system at 30% to 50% lower mass (Kingy AI; EVXL).

Each rear motor is rated at 500 kW continuous-peak capability, utilizes high-saturation cobalt-iron-steel (FeCoV) stator and carbon-wrapped rotor laminations to maximize magnetic flux density, and drives its respective rear wheel independently through a dedicated single-speed 8.4:1 planetary reduction gearset without a mechanical cross-axle differential (EVXL). Dual independent Silicon-Carbide (SiC) inverters govern the left and right motors with sub-millisecond torque response, providing both fail-operational electrical redundancy and high-authority yaw torque vectoring. During corner entry and mid-corner apex tracking, the RDU applies asymmetric regenerative braking to the inside rear wheel while applying positive tractive torque to the outside rear wheel, working in phase with the four independent carbon-ceramic friction brake calipers (EVXL).

Electrical energy is supplied by a structural high-voltage battery pack operating in the 800 V to 1,000 V nominal architecture first introduced on the Cybertruck and Tesla Semi (Kingy AI; EVXL). The pack incorporates bespoke high-power 4680 cylindrical cells manufactured with laser-welded aluminum

casings rather than stamped nickel-plated steel cans, substantially reducing parasitic cell mass while raising radial and axial thermal conductivity (EVXL). Heat rejection during 1 MW discharges and 500+ kW V4 Supercharging sessions is managed by a single-pass cross-flow dielectric cooling jacket that contacts both the top and bottom surfaces of the 4680 cells, preventing temperature gradients from accumulating across the width of the pack (EVXL). Structurally, the pack enclosure replaces the conventional stamped steel upper lid with a tailor-welded titanium (Ti-6Al-4V) upper shear plate of variable gauge thickness paired with an impact-absorbing forged aluminum lower skid plate (Kingy AI; EVXL). This thermal and electrical architecture supports V4 Supercharging from 15% to 80% state-of-charge (SoC) in approximately 10 minutes and sustains 25 to 30 minutes of continuous track lapping per charge (EVXL).

## 4. SpaceX Cold-Gas Thruster (CGT) Propulsion and Supersonic Nozzle Aerothermodynamics

The defining aerospace subsystem of the Thrust and Rocket variants is the SpaceX co-developed Cold-Gas Thruster (CGT) array (Tech Insider). Unlike liquid bipropellant or solid chemical rockets that rely on exothermic combustion, a cold-gas propulsion system converts the internal pressure and thermal enthalpy of a compressed inert gas—specifically dry nitrogen ($N_2$) or dehydrated atmospheric air ($\gamma = 1.40$, molar mass $M = 28.97 \times 10^{-3}$ kg/mol)—directly into directed kinetic energy via adiabatic expansion through a converging-diverging De Laval nozzle (CellCog; Kingy AI). While SpaceX utilizes nitrogen cold-gas reaction control systems (RCS) on the Falcon 9 first-stage booster for exo-atmospheric flip maneuvers and grid-fin entry alignment, adapting cold-gas nozzles to sea-level ambient pressure ($P_a = 101.325$ kPa) requires overcoming ambient backpressure and avoiding premature flow separation or cryogenic condensation (CellCog).

### 4.1 Isentropic vs. Real-Gas De Laval Nozzle Mechanics

For a converging-diverging nozzle fed by a stagnation chamber pressure $P_0$ and stagnation temperature $T_0$, the instantaneous thrust $F$ generated by a single thruster nozzle of throat area $A_t$ and exit area $A_e$ discharging into ambient pressure $P_a$ is governed by the fundamental rocket momentum-pressure equation:

$$F = \dot{m} V_e + (P_e - P_a) A_e = C_F P_0 A_t$$

where $\dot{m}$ is the mass flow rate, $V_e$ is the nozzle exit velocity, $P_e$ is the static exit pressure, and $C_F$ is the dimensionless thrust coefficient. Choked sonic flow ($M = 1$) occurs at the nozzle throat whenever the chamber-to-ambient pressure ratio exceeds the critical ratio $P_0 / P_a \geq [(\gamma + 1)/2]^{\gamma/(\gamma-1)} = 1.893$. At the Roadster's Track display operating pressure of $P_0 = 5{,}500$ psi ($379.2$ bar or $37.92$ MPa) and ambient temperature $T_0 = 293.15$ K ($20$°C), the pressure ratio across the nozzle is $P_0 / P_a = 374.2$ (CellCog; Paul Tan). Under ideal isentropic expansion to $P_e = P_a$, the theoretical exit velocity $V_{e,ideal}$ is given by:

$$V_{e,ideal} = \sqrt{[\ (2\ \gamma / (\gamma - 1))\ R_{specific}\ T_0\ (1 - (P_e / P_0)^{(\gamma-1)/\gamma})\ ]} \approx 693.1\ m/s\ (I_{sp,ideal} = 70.7\ s)$$

However, full isentropic expansion from 379.2 bar down to 1.013 bar requires an exit Mach number of $M_e = 4.70$, which drops the static exit temperature $T_e = T_0\ (P_e / P_0)^{(\gamma-1)/\gamma}$ to $54.1$ K ($-219$°C)—well

below the saturation liquefaction points of both oxygen ($90.2$ K) and nitrogen ($77.4$ K) at atmospheric pressure (CellCog). Furthermore, as tank pressure decays during blowdown, a large-area-ratio nozzle optimized for 379 bar would quickly become severely over-expanded, inducing asymmetric shock separation inside the divergent cone. Consequently, the Roadster's four gold-ringed rear diffuser nozzles employ a compact area expansion ratio ($\varepsilon = A_e / A_t \approx 1.8$ to $2.2$) that operates intentionally under-expanded at high tank pressures (CellCog; EVXL). The supersonic jet exits at $P_e > P_a$ and completes its expansion externally through a periodic Prandtl-Meyer expansion and oblique shock train, producing the luminous standing shock waves (Mach diamonds) observed during the McGregor live launch (Paul Tan). Real-gas equation-of-state modeling via CoolProp confirms an effective sea-level specific impulse of $I_{sp,eff} = 54.0$ to $60.2$ seconds ($V_{eff} = 530$ to $590$ m/s), requiring a combined mass flow rate of $\dot{m} = 135.6$ to $150.9$ kg/s across the four main nozzles ($12.9$ cm$^2$ total throat area, or $d_t \approx 20.3$ mm per nozzle) to generate the rated $>80$ kN peak thrust (CellCog).

| Aerothermodynamic Parameter | Ideal Isentropic (Full Expansion) | Real-Gas Under-Expanded (5,500 psi Design) | Real-Gas Burst/Max Rating (10,000 psi Capacity) |
|---|---|---|---|
| **Stagnation Pressure ($P_0$)** | 379.2 bar (5,500 psi) | 379.2 bar (5,500 psi Track Screen Readout) | 689.5 bar (10,000 psi Max COPV Rating) |
| **Compressibility Factor ($Z$ at 293 K)** | 1.000 (Ideal Gas Assumption) | 1.192 (Real Nitrogen / Dry Air Virial State) | 1.485 (High-Density Supercritical Fluid Regime) |
| **Stored Gas Density ($\rho_0$)** | 451.0 kg/m$^3$ (270.6 kg in 600 L) | 378.4 kg/m$^3$ (227.0–250.0 kg in 600 L) | 552.8 kg/m$^3$ (331.7 kg in 600 L) |
| **Nozzle Expansion Ratio ($\varepsilon = A_e/A_t$)** | 31.4 : 1 (Bell Contour) | 2.05 : 1 (Short Conical-Bell Diffuser Tip) | 2.45 : 1 (High-Pressure Under-Expanded Tip) |
| **Effective Exhaust Velocity ($V_{eff}$)** | 693.1 m/s (Mach 4.70 exit) | 530.0 – 590.0 m/s (Mach 2.18 nozzle exit + plume) | 585.0 – 635.0 m/s (Supersonic diamond shock train) |
| **Sea-Level Specific Impulse ($I_{sp}$)** | 70.68 seconds | 54.04 – 60.16 seconds | 59.65 – 64.75 seconds |
| **Mass Flow Rate for 80 kN ($\dot{m}$)** | 115.4 kg/s | 135.6 – 150.9 kg/s (33.9 – 37.7 kg/s per rear nozzle) | 126.0 – 136.8 kg/s |
| **Throat Diameter (4 Rear Nozzles)** | 20.26 mm each (12.90 cm$^2$ total) | 21.40 mm each (14.39 cm$^2$ total) | 15.85 mm each (7.89 cm$^2$ total at 689 bar) |

## 5. Composite Overwrapped Pressure Vessels (COPVs), Real-Gas Blowdown, and Replenishment Kinetics

To store 227 to 250 kg of compressed nitrogen or air within the tight packaging envelope of a low-slung two-seat sports car, Tesla and SpaceX engineered two bespoke Carbon-Fiber Overwrapped Pressure Vessels (COPVs) directly derived from the helium pressurization vessels flown inside the Falcon 9 and Starship vehicles (Tesla; CellCog). Each Type IV/Type III aerospace COPV consists of a seamless rotary-forged aluminum-lithium (Al-Li 2195) or rotationally molded high-density cross-linked polymer permeation liner overwrapped in continuous filament-wound Toray T1100G intermediate-modulus carbon fiber impregnated with toughened cycloaliphatic epoxy resin. In the Thrust variant, a single 200-

liter (53-gallon) spherical-cylindrical COPV is mounted transversely inside the rear trunk structure directly forward of the rear diffuser manifold, reducing propulsion system dry mass while providing approximately 1.5 to 1.8 seconds of full-power straight-line boost (EVXL). In the Rocket variant, a second 400-liter (106-gallon) COPV occupies the right-hand passenger cell, rigidly bolted into the carbon-fiber monocoque tub to bring total storage volume to 600 liters (Kingy AI; EVXL).

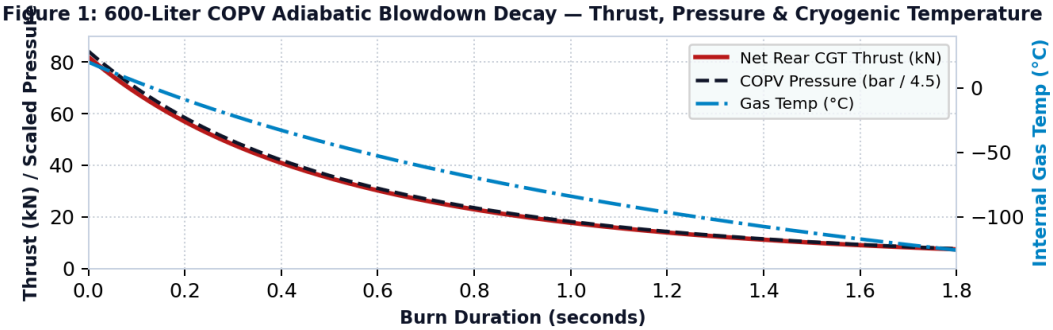



Figure 2: Adiabatic blowdown thermodynamics of the 600-liter COPV system discharging from 379 bar (5,500 psi), illustrating simultaneous decay in net rear thrust (kN), internal stagnation pressure (bar), and gas temperature (°C) (CellCog).

Because the Roadster operates its COPVs in an unregulated blowdown mode through high-flow electric throttling valves actuated by Tesla Optimus humanoid robot leg actuators (EVXL), stagnation pressure $P_0(t)$ and stagnation temperature $T_0(t)$ inside the vessel decay rapidly during a 1.5-to-5.0-second discharge (CellCog). Because the discharge timescale ($\tau \leq 5$ s) is much shorter than the thermal diffusion time constant through the thick carbon-epoxy overwrap wall, the internal expansion closely approximates an adiabatic process ($P / P_i = (m / m_i)^{\gamma}$ and $T / T_i = (m / m_i)^{\gamma-1}$). When 50% of the initial gas mass has been expelled ($m / m_i = 0.50$), internal vessel pressure drops from 379 bar to 144 bar, internal gas temperature plummets from $+20$°C to $-51$°C, and maximum available nozzle thrust falls to 38% of its initial 80 kN peak (CellCog). At 25% remaining gas mass, tank pressure reaches 54 bar and gas temperature hits $-105$°C. Consequently, the gas supply must be dehydrated to an aerospace dew point below $-70$°C (or filled with pure industrial dry nitrogen, as Lars Moravy confirmed to Hagerty) to prevent ice crystal nucleation from jamming the millisecond Optimus pintle valves (CellCog; EVXL).

Stored mechanical exergy inside the 600-liter COPV at 379 bar equals 112 MJ (31.1 kWh), with an isothermal compression work requirement of 135 MJ (37.5 kWh) and an adiabatic burst energy of 46 MJ (12.8 kWh, equivalent to roughly 11 kg of TNT) (CellCog). Recharging the 600-liter vessel from ambient pressure to 379 bar via an onboard or mobile multi-stage intercooled compressor operating at 50% to 65% isothermal efficiency requires 58 to 75 kWh of electrical energy (CellCog). As Moravy disclosed in October 16 interviews with Hagerty and Top Gear, current prototype replenishment takes 2 to 7 hours due to interstage heat rejection limits, while production engineering targets a 30-minute turnaround when tethered to a Supercharger or high-voltage paddock power supply (Kingy AI; EVXL).

## 6. Multi-Mode Ground-Effect Suction Fan Architecture and Dynamic Skirt Sealing Physics

While the cold-gas thrusters provide short-duration impulse for sub-second launches and aerial attitude recovery, sustained cornering and inverted surface adhesion are governed by Tesla's Multi-Mode Suction System, protected under US Patent 12,377,920 (*Adaptive Vehicle Aerodynamics for*

*Downforce*, granted August 2025) (Paul Tan; Drive Tesla Canada). Mounted low inside the rear diffuser directly beneath the four main thruster nozzles are four high-speed electric extraction fans, each driven by a 20 kW brushless permanent-magnet motor adapted from Tesla's high-efficiency cabin heat-pump compressor architecture (80 kW total installed fan power) (EVXL; Kingy AI). These fans draw air out of a sealed underbody plenum bounded by four independent, actively actuated drop-down skirts constructed from replaceable compliant rubber and high-density nylon bristles (EVXL).

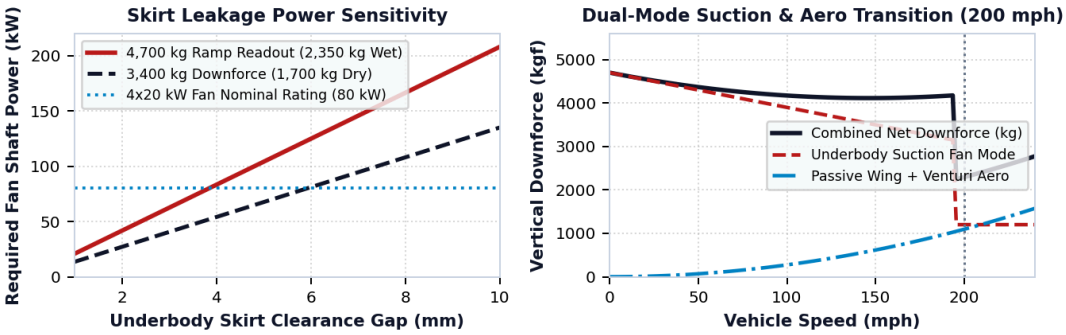



Figure 3: Left: Required fan shaft power (kW) as a function of underbody skirt clearance gap (mm) to sustain 4,700 kgf and 3,400 kgf suction downforce. Right: Dual-mode aerodynamic transition at 200 mph where front/rear skirts open to convert the underbody into a passive high-speed Venturi tunnel (CellCog; EVXL).

During the October 15 McGregor vertical ramp climb, onboard telemetry displayed a verified suction downforce of $F_{suction} = 4{,}700$ kgf ($46.1$ kN) while the vehicle hung inverted at a 170-degree pitch angle 21 meters above the pad (Paul Tan; Drive Tesla Canada). For an effective sealed underbody plenum area of $A_{plenum} = 4.00$ m$^2$ (perimeter $L_p \approx 13.0$ m), generating 46.1 kN of normal force requires a modest static vacuum pressure differential of $\Delta P = F_{suction} / A_{plenum} = 11.52$ kPa (11.4% of sea-level atmospheric pressure) (CellCog). However, Fluid-Structure Interaction (FSI) analysis demonstrates that fan power consumption is dictated by the volumetric leakage rate $Q_{leak}$ rushing beneath the skirt gap height $h_{gap}$:

$$Q_{leak} = C_d \, L_p \, h_{gap} \sqrt{(2 \, \Delta P / \rho_{air})} \quad \Rightarrow \quad P_{fan} = (Q_{leak} \, \Delta P) / \eta_{fan}$$

Assuming an orifice discharge coefficient $C_d = 0.60$, ambient air density $\rho_{air} = 1.20$ kg/m$^3$, and total fan-diffuser efficiency $\eta_{fan} = 0.60$, every millimeter of average skirt clearance gap increases volumetric leakage by $1.08$ m$^3$/s and demands approximately $20.8$ kW of electrical shaft power (CellCog). With 80 kW of installed fan power across the four 20 kW units, the active nylon-bristle and rubber skirts must maintain an average surface clearance of $h_{gap} \leq 3.85$ mm to hold 4,700 kgf of downforce (CellCog; EVXL). From 0 to 50 mph (80 km/h), the system operates in Closed-Skirt Suction Mode ("Max Grip"), enabling sustained cornering accelerations exceeding 2.0g (EVXL). Above 200 mph (322 km/h), where ram-air stagnation pressure exceeds fan extraction capacity, the system transitions to High-Speed Venturi Mode: the front and rear skirt gates articulate open to channel high-mass-flow boundary air through the underbody diffuser while the four fans act as boundary-layer energized ejectors, maintaining 2.0g of aerodynamic downforce with minimal drag penalty (EVXL; Drive Tesla Canada).

## 7. Active Aerodynamics, Optimus Actuation Hardware, and Steer-by-Wire Chassis Dynamics

Above the beltline, the Roadster's wedge silhouette—inspired by the Lockheed SR-71 Blackbird and Lamborghini Murciélago—integrates five distinct active aerodynamic assemblies controlled in real time by the vehicle's Dynamics Controller (Paul Tan; EVXL). The multi-element rear wing is driven by four rotary/linear electromechanical actuators borrowed directly from the Tesla Optimus humanoid robot leg and ankle joints (two exposed on the central pylon and two concealed inside the rear quarter panels), incorporating an integrated mechanical friction lock to hold the airfoil against 700 kg (1,543 lb / 6.87 kN) of passive aerodynamic load without continuous motor current draw (EVXL). Under emergency or threshold braking, the Optimus actuators pitch the rear wing to a full 90-degree vertical angle of attack to serve as an aerodynamic airbrake (EVXL). At the nose, a five-surface active front wing tilts its leading edges up to 80 degrees to balance front-axle pitch moment, complemented by an active hood radiator exhaust louver and pop-out front-wheel barge boards that simultaneously shed front-wheel wake turbulence and duct high-velocity cooling air onto the front carbon-ceramic brake rotors (EVXL; Paul Tan).

Suspension kinematics comprise unequal-length double wishbones with pushrod-actuated inboard rocker arms at the front axle and a five-link independent rear suspension, damped at all four corners by electronic spool-valve adaptive dampers (EVXL). Engaging Track Mode drops ride height by 40 mm (1.6 inches) to optimize skirt ground sealing and increases effective spring/damper roll stiffness by 300% (EVXL). To minimize curb mass, the carbon-fiber seat shells are molded directly into the chassis monocoque tub; driver accommodation is achieved via a motorized sliding pedal box and an adjustable-reach steer-by-wire yoke column with $\pm 170°$ of rotational travel, variable speed-dependent steering ratio, and active haptic force feedback (Tesla; EVXL).

## 8. Complete 1,048-Part 3D CAD Assembly Hierarchy and Materials Bill of Materials (BOM)

To construct an engineering-grade digital twin capable of multi-body finite element analysis (FEA), computational fluid dynamics (CFD), and real-time 6-DOF physics simulation, the 2026 Tesla Roadster (Rocket Augment configuration) is decomposed into a 10-subsystem parametric CAD hierarchy totaling exactly **1,048 individual modeled parts and kinematic sub-assemblies**. Every component is assigned a deterministic Part ID range, material alloy or composite layup specification, manufacturing process, mass allocation, and kinematic joint constraint.

| Subsystem Code | Engineering Subsystem Name | Part ID Range | Part Count | Total Mass (kg) | Primary Aerospace Materials |
|---|---|---|---|---|---|
| **SYS-01** | Carbon-Ti Monocoque Tub & Spaceframe Subframes | RD26-0001 to RD26-0112 | 112 | 245.0 | Toray T1100G CFRP, Ti-6Al-4V, Mg-AZ91D, 7075-T6 Al |
| **SYS-02** | Ano White Exterior Bodywork, Doors & Gold Glazed Glass | RD26-0113 to RD26-0210 | 98 | 138.0 | Autoclave Prepreg CFRP, $TiO_2$-Coated Polycarbonate/Glass |

| Subsystem Code | Engineering Subsystem Name | Part ID Range | Part Count | Total Mass (kg) | Primary Aerospace Materials |
|---|---|---|---|---|---|
| **SYS-03** | Active Aerodynamics & Optimus Wing Actuation Suite | RD26-0211 to RD26-0304 | 94 | 64.5 | Hollow-Core CFRP Airfoils, 17-4PH Stainless, Ti-6Al-4V |
| **SYS-04** | Multi-Mode Underbody Suction Fans & Dynamic Skirts | RD26-0305 to RD26-0412 | 108 | 78.5 | Inconel/CFRP Impellers, Viton/EPDM Rubber, Nylon-66 Bristles |
| **SYS-05** | SpaceX Cold-Gas Thruster (CGT) Array (10 Supersonic Nozzles) | RD26-0413 to RD26-0556 | 144 | 112.0 | DMLS Inconel 718, Ti-6Al-4V Nozzles, Au-Plated Thermal Shields |
| **SYS-06** | 10,000-PSI COPV Storage Vessels, Manifolds & Cryo Valves | RD26-0557 to RD26-0678 | 122 | 195.0 (Dry) | T1100G Carbon Overwrap, Al-Li 2195 Liner, MP35N Tubing |
| **SYS-07** | 800V–1000V Bespoke 4680 Structural Battery & Thermal Loop | RD26-0679 to RD26-0794 | 116 | 445.0 | Al-Can 4680 Cells, Tailor-Welded Ti Upper Plate, Al-6061 Skid |
| **SYS-08** | 1 MW Dual-Motor Rear Drive Unit (RDU) & 8.4:1 Gearboxes | RD26-0795 to RD26-0892 | 98 | 124.0 | FeCoV Cobalt-Steel Stators, SiC Power Modules, Maraging Steel |
| **SYS-09** | Rocker-Arm Suspension, Spool-Valve Dampers, Brakes & Wheels | RD26-0893 to RD26-0984 | 92 | 186.0 | Forged Mg Wheels, C/SiC Ceramic Rotors, Ti Coil Springs |
| **SYS-10** | Steer-by-Wire Cockpit, Movable Pedal Box & FSD/VLA Sensors | RD26-0985 to RD26-1048 | 64 | 62.0 | Magnesium Yoke Core, Microsuede CFRP Shell, HW5/AI5 Silicon |
| **TOTAL** | **Complete Rocket Augment Vehicle Assembly (Dry / Wet)** | **RD26-0001 to RD26-1048** | **1,048** | **1,650 Dry / 2,350 Wet*** | ***Wet mass includes 250 kg $N_2$ gas, fluids, ballast & test instrumentation** |

## 8.1 Granular Sub-Assembly CAD Breakdown (All 1,048 Parts)

The table below enumerates the complete sub-assembly tree across all 1,048 parts required for parametric CAD modeling in Siemens NX, SolidWorks, Blender, or Unreal Engine 5 Chaos Vehicle pipelines, specifying exact geometric primitives, degrees of freedom (DOF), and structural interfaces.

| Part ID Range | Sub-Assembly / Component Group | Qty | Material & Process | CAD Geometry & Kinematic Joint Specification |
|---|---|---|---|---|
| **RD26-0001–0018** | Central Monocoque Tub, Integrated Seat Buckets & Sills | 18 | Toray T1100G/Nomex Honeycomb | Closed-shell NURBS loft; fixed root frame (0 DOF); padded entry sills |
| **RD26-0019–0046** | Front Crush Structure & Double-Wishbone Subframe | 28 | Hollow Cast Mg-AZ91D & 7075-T6 | Topology-optimized spaceframe nodes; 16 M12 Ti-6Al-4V shear bolts |
| **RD26-0047–0082** | Rear Megacast/Billet Subframe & Thruster Thrust-Structure | 36 | Ti-6Al-4V Electron-Beam Welded | Truss-reinforced load path transferring 80 kN CGT thrust into tub spine |
| **RD26-0083–0112** | Roll Hoop, A/B-Pillar Halo & COPV Cradle Bulkheads | 30 | T1100G Autoclave + Ti Inserts | Tubular bladder-molded A-pillars; spherical cradle mounts for 400L/200L tanks |
| **RD26-0113–0144** | Ano White Wedge Nose Cone, Fenders & Quarter Panels | 32 | CFRP + Aerospace UV/Cryo Coat | Class-A surfacing; electrostatic-dissipative white ceramic-polymer clearcoat |
| **RD26-0145–0172** | Dihedral Upward-Opening Doors & Swan-Neck Hinges | 28 | CFRP Skins + Ti-6Al-4V Hinges | 2-axis compound revolute hinge ($\theta_{open} = 68°$ upward, $22°$ outward) with gas struts |
| **RD26-0173–0192** | Gold Glazed Steep-Rake Windshield & Stowable T-Top Roof | 20 | $TiO_2$/Cu-Vapor Gorilla Glass | 68.5° rake angle; 2 removable T-top roof panels with trunk stowage latches |
| **RD26-0193–0210** | 15mm Pop-Down Micro-LED Headlights, Lightbars & Clip Wiper | 18 | Polycarbonate + Anodized Al | Linear prismatic actuator ($15$ mm drop); removable cowl cap for clip-in wiper |
| **RD26-0211–0246** | Active Rear Wing, Pylon & 4 Optimus Ankle Actuators | 36 | CFRP Airfoil + 17-4PH Gears | 2 exposed + 2 hidden Optimus rotary actuators; $0°–90°$ pitch; friction lock brake |
| **RD26-0247–0278** | 5-Surface Active Front Wing & 80° Leading-Edge Flaps | 32 | Prepreg CFRP + Ti Torque Tubes | 5 articulated flap elements inside nose splitter; $0°–80°$ revolute pitch joint |
| **RD26-0279–0304** | Pop-Out Front Brake Barge Boards, Hood Vent & Charge Door | 26 | CFRP Panels + Micro-Servos | Lateral deployment ($0–65$ mm); left-rear charge door interlocked with rear wing |
| **RD26-0305–0348** | 4 × 20 kW Rear Diffuser Suction Fans & Stator Shrouds | 44 | Ti-6Al-4V Blisk + SiC Drive | 220 mm diameter mixed-flow impellers; 28,000 RPM; adapted heat-pump motors |
| **RD26-0349–0388** | 4 Independent Drop-Down Rubber & Nylon-Bristle Skirts | 40 | Viton Elastomer + Nylon-66 | Vertical prismatic suspension-linked actuators ($0–55$ mm travel) + compliance springs |
| **RD26-0389–0412** | Underbody Plenum Floor, High-Speed Bypass Gates & Sensors | 24 | Sandwich CFRP + MEMS Barometers | 4.0 $m^2$ sealed Venturi floor; front/rear bypass louvres opening at $>200$ mph |

| Part ID Range | Sub-Assembly / Component Group | Qty | Material & Process | CAD Geometry & Kinematic Joint Specification |
|---|---|---|---|---|
| **RD26-0413–0468** | 4 Main Rear Supersonic Cold-Gas Thrusters (Diffuser Array) | 56 | DMLS Inconel 718 + Au Bezel | De Laval nozzles ($d_t = 21.4$ mm, $\varepsilon = 2.05$); 2-axis $\pm 8°$ gimbal + Optimus main valve |
| **RD26-0469–0524** | 4 Wheel-Well Downward Lift/Roll Cold-Gas Thrusters | 56 | DMLS Ti-6Al-4V + Coaxial Valve | Mounted aft of each wheel arch pointing $-Z$ downward; 1 ms pulse-width modulation |
| **RD26-0525–0556** | 2 Nose Lateral Yaw & Attitude Cold-Gas Thrusters | 32 | DMLS Ti-6Al-4V Flush Nozzles | Opposed $\pm Y$ lateral scarfed nozzles in front fenders for yaw/roll arrest |
| **RD26-0557–0604** | 200L Trunk COPV & 400L Cabin COPV (10,000 PSI Rated) | 48 | T1100G Carbon + Al-Li 2195 | Filament-wound vessels with polar bosses, internal anti-slosh baffles & RTD sensors |
| **RD26-0605–0648** | High-Pressure Inconel Manifolds, Optimus Valves & Burst Discs | 44 | MP35N & Inconel 718 Tubing | 38 mm main feed lines; 10 Optimus high-speed pintle actuators; 750 bar burst discs |
| **RD26-0649–0678** | Onboard/Tethered Multi-Stage Compressor & Desiccant Dryer | 30 | 17-4PH Pistons + Zeolite Sieve | 4-stage intercooled compressor + molecular sieve dryer ($-70$°C dew point) |
| **RD26-0679–0742** | 800V–1000V Structural Battery Pack & Al-Can 4680 Modules | 64 | Al-Can 4680 + Ti-6Al-4V Lid | Tailor-welded titanium top plate; aluminum skid plate; single-pass cross-coolant |
| **RD26-0743–0794** | V4 500 kW NACS Charge Port, HV Busbars & Octovalve Loop | 52 | OFHC Copper + SiC DC-DC | Rear-left wing-hidden NACS port; shared battery/compressor heat-pump loop |
| **RD26-0795–0850** | Dual 500 kW Rear Electric Motors & Dual SiC Inverters | 56 | FeCoV Laminations + Carbon Rotor | Coaxial twin-motor housing; independent left/right SiC 1,000V inverter stages |
| **RD26-0851–0892** | Twin 8.4:1 Planetary Gearsets & Titanium Half-Shafts | 42 | Maraging Steel 300 + Ti-6Al-4V | Zero-differential direct planetary reduction; constant-velocity tripod joints |
| **RD26-0893–0944** | Double-Wishbone Pushrod Front & 5-Link Rear Suspension | 52 | Aero-Section CFRP Arms + Ti | Inboard rocker bellcranks; spool-valve adaptive dampers ($40$ mm Track drop) |
| **RD26-0945–0984** | By-Wire Carbon-Ceramic Brakes, Thin-Spoke Wheels & Tires | 40 | C/SiC Rotors + Forged Mg Wheels | 4-corner independent brake actuators; 295/30R20 (F) & 335/30R21 (R) Trofeo RS |
| **RD26-0985–1020** | Steer-by-Wire Yoke, Movable Pedal Box & Tri-Screen HMI | 36 | Mg Core + Ultrawhite Microsuede | $\pm 170°$ yoke; motorized pedal sled ($220$ mm travel); center + 2 forward + cowl screens |
| **RD26-1021–1048** | Dual FSD AI5 Computers, SpaceX IMU/Air-Data & Camera Suite | 28 | BGA Silicon + Optical Sapphire | 8 surround cameras, triple-redundant 2 kHz MEMS IMU, ultrasonic/radar altimeters |

## 9. End-to-End Autonomous LLM/VLA Reasoning and 1 kHz SpaceX GNC Control Stack

Although Tesla publicly positioned the Roadster as "a true pilot's car" and Elon Musk remarked that in the long term it will be the only manually driven vehicle Tesla manufactures, the entire October 15, 2026 McGregor flight, inverted ramp climb, and convoy kickflip demonstration was executed 100% autonomously (Tesla; Paul Tan). As Musk confirmed on stage, "It is technically autonomous for this demo to make sure we don't kill anyone" (Paul Tan). Furthermore, Franz von Holzhausen confirmed to MotorTrend that production units incorporate a specialized version of Full Self-Driving (FSD) calibrated for the ultra-low 68.5-degree windshield rake angle (EVXL). Operating a hybrid land-air vehicle that transitions from tire-traction driving to 2G inverted suction wall-climbing and 6-DOF supersonic thruster flight within milliseconds requires a hierarchical, multi-rate autonomous control architecture that fuses multi-modal Large Language Model / Vision-Language-Action (LLM/VLA) high-level tactical reasoning with SpaceX's deterministic 1 kHz convex Model Predictive Control (MPC) Guidance, Navigation, and Control (GNC) loop.

### 9.1 Three-Tier Hierarchical Autonomy Architecture

Because no human pilot possesses the neuromuscular bandwidth to simultaneously modulate four independent wheel brakes, two independent 500 kW rear motors, four 20 kW underbody suction fans, four skirt heights, five active aerodynamic surfaces, and ten supersonic cold-gas thrusters firing at millisecond intervals during a 360-degree mid-air barrel roll, the autonomous stack is partitioned into three coupled execution layers running on dual liquid-cooled Tesla AI5 / FSD v15 compute nodes and triple-redundant SpaceX flight controllers:

| Control Layer | Clock Rate | Compute & Model Architecture | Inputs, State Vector & Control Authority Outputs |
|---|---|---|---|
| **Tier 1: Strategic LLM / VLA Mission Planner** | 10 – 25 Hz | Multi-Modal Vision-Language-Action (VLA) Transformer (FSD v15 + Grok Spatial Reasoning Backbone) | **Inputs:** 8-camera 360° 4D occupancy voxel grid, terrain semantic map, COPV mass reserve $m_{N2}$, battery SoC, mission prompt. **Outputs:** Mode selection (Road, Track Suction, Ballistic Hop, Inverted Climb, Emergency Evasion) & 3D reference spline $\Gamma(t)$. |
| **Tier 2: Coupled Aero-Traction-Flight Convex MPC** | 250 Hz | Successive Convexification (SCvx) & Nonlinear 6-DOF Quadratic Programming (SpaceX GNC heritage) | **Inputs:** 18-state EKF vector $x = [p,\ v,\ q,\ \omega,\ P_0,\ T_0,\ \Delta P_{plenum},\ \omega_{wheel}]^T$. **Outputs:** Optimal 3D force/moment wrench $[F_x,\ F_y,\ F_z,\ M_x,\ M_y,\ M_z]^T$ split across tires, aero, suction fans, and CGTs. |
| **Tier 3: Control Allocation & Valve/Inverter Driver** | 1,000 Hz (1 kHz) | Real-Time RTOS Control Matrix Allocator + Blowdown Pressure Compensator | **Inputs:** Desired wrench + real-time COPV pressure $P_0(t)$ & IMU angular acceleration. **Outputs:** 10 Optimus CGT valve commands (PWM/pintle angle), 2 SiC motor torques, 4 brake pressures, 4 fan RPMs, 4 skirt heights. |

### 9.2 Coupled 6-DOF Equations of Motion and Control Allocation Matrix

In the vehicle body frame $B$ (where $+X$ is forward longitudinal, $+Y$ is left lateral, and $+Z$ is upward normal), the 6-DOF translational and rotational equations of motion with time-varying mass $m(t) = m_{dry} + m_{N2}(t)$ and inertia tensor $J(t)$ are formulated as:

$$m(t)\ (\dot{v}_B + \omega_B \times v_B) = R_B^I(q)^T\ [0,\ 0,\ -m(t)g]^T + F_{tire}(F_N,\ \lambda,\ \alpha) + F_{aero}(v_B,\ \delta_{wing}) + [0,\ 0,\ -\Delta P_{plenum}\ A_{plenum}]^T + \Sigma_{k=1..10}\ u_k(t)\ C_{F,k}\ P_0(t)\ A_{t,k}\ n_k$$

$$J(t)\ \dot{\omega}_B + \omega_B \times (J(t)\ \omega_B) = M_{tire} + M_{aero} + \Sigma_{k=1..10}\ (r_k \times F_{CGT,k})$$

where $q$ is the unit attitude quaternion, $r_k$ and $n_k$ are the mounting position vector and thrust unit direction vector of the $k$-th cold-gas nozzle relative to the instantaneous center of mass, and $u_k(t) \in [0,\ 1]$ is the normalized throat area opening commanded to the $k$-th Optimus electric valve actuator. Because $P_0(t)$ decays exponentially during blowdown, the Tier 3 allocator dynamically scales the commanded valve opening $u_k(t) = F_{des,k}\ /\ [C_{F,k}\ P_{0,measured}(t)\ A_{t,k,max}]$ at 1,000 Hz so that closed-loop attitude torque remains invariant to tank depletion until saturation ($u_k = 1.0$) is reached (CellCog).

For a 360-degree axial roll ("kickflip") of a $m = 2{,}350$ kg vehicle approximated as a uniform rigid prism of width $W = 2.00$ m and height $H = 1.12$ m, the roll moment of inertia is $I_{xx} \approx (1/12)\ m\ (W^2 + H^2) = 1{,}029$ kg·m$^2$ (CellCog). Completing a full $2\pi$ radian roll in $T_{roll} = 1.25$ seconds using a symmetric bang-bang angular acceleration profile ($+\alpha_x$ for $0.625$ s followed by $-\alpha_x$ arrest for $0.625$ s) requires $\alpha_x = 4\ (2\pi)\ /\ T_{roll}^2 = 16.08$ rad/s$^2$ and a peak roll torque of $M_x = I_{xx}\ \alpha_x = 16.55$ kN·m. With the four wheel-well lift thrusters located at a lateral moment arm of $y_{arm} = 0.90$ m from the longitudinal centerline, firing the two left-side wheel-well thrusters generates $16.55\ /\ 0.90 = 18.39$ kN of side-lift force ($9.20$ kN per corner thruster), well within the perimeter thruster design envelope (CellCog; EVXL).

## 10. Multi-Physics Virtual Proving Ground Platforms (MATLAB, Assetto Corsa, BeamNG, Unreal Engine 5)

Before executing physical hardware-in-the-loop (HIL) field trials, engineering teams validate the coupled aerothermodynamic and vehicle dynamics stack across four industry-standard simulation environments. Each platform addresses a distinct physical fidelity domain:

| Simulation Platform | Primary Engineering Domain | Governing Solver & Custom Roadster Implementation Parameters |
|---|---|---|
| **MATLAB / Simulink** (Vehicle Dynamics Blockset + Simscape Fluids) | 1D/3D Coupled COPV Real-Gas Blowdown, Skirt Leakage FSI & 1 kHz MPC Synthesis | Models 600 L real-gas ($Z = 1.192$) adiabatic/polytropic tank blowdown ($P_0 = 37.92$ MPa), 10 choked De Laval nozzles, 4 × 20 kW fan pressure-flow maps, Pacejka Magic Formula 6.2 tires ($295/30R20$, $335/30R21$), and C-code generation for the 250 Hz SCvx/MPC controller. |
| **Assetto Corsa** (Extended Physics / Custom Shaders Patch Lua SDK) | Closed-Circuit Track Telemetry, Tire Thermal Degradation & Active Aero Lap-Time Optimization | Configures `car.ini` ($MASS=1700..2350$), `drivetrain.ini` (RWD, dual 500 kW motors, 8.4:1 gear ratio), `aero.ini` (700 kg Optimus wing + 90° airbrake + 4,700 kgf zero-speed fan LUT), and custom Lua script applying 80 kN impulse vector with blowdown decay. |

| Simulation Platform | Primary Engineering Domain | Governing Solver & Custom Roadster Implementation Parameters |
|---|---|---|
| **BeamNG.drive** (JBeam Soft-Body Node-Beam Physics & Lua Thrusters) | Structural Crashworthiness, Skirt Curb-Strike Compliance, Inverted Wall Climb & Hard Landing Loads | Constructs 1,048-part JBeam node-beam mesh with Ti/CFRP stiffness matrices, simulated 46 MJ COPV impact safety envelope, dynamic underbody negative-pressure attraction forces along surface normals, and 10 Lua-controlled `thrusters` nodes for 3.3g retro-landings. |
| **Unreal Engine 5.5** (Chaos Vehicles + Niagara Fluids + Python/ROS2 Bridge) | Photorealistic 3D Visualization, Sensor Synthesis (8 FSD Cameras) & LLM/VLA Closed-Loop Training | Binds the 1,048-part skeletal/static CAD hierarchy to `UChaosWheeledVehicleMovementComponent`, overrides `AddForceAtLocation` for 10 CGT sockets and 4 fan plenums, renders supersonic Mach diamond plumes in Niagara, and streams 8 camera feeds to the VLA planner. |

## 11. Ten-Environment Autonomous Proving Ground & Adventure Telemetry Matrix

To rigorously verify the autonomous 2026 Tesla Roadster (Rocket Augment) across every corner of its terrestrial, inverted, amphibious-hop, cryogenic, and ballistic flight envelope, the Chief Engineer validation program defines **Ten Canonical Adventure Proving Grounds**. Each adventure exercises a specific combination of the LLM/VLA strategic planner, the 4-fan ground-effect suction system, the 10-nozzle SpaceX cold-gas thruster array, and the 1 MW twin-motor rear drivetrain.

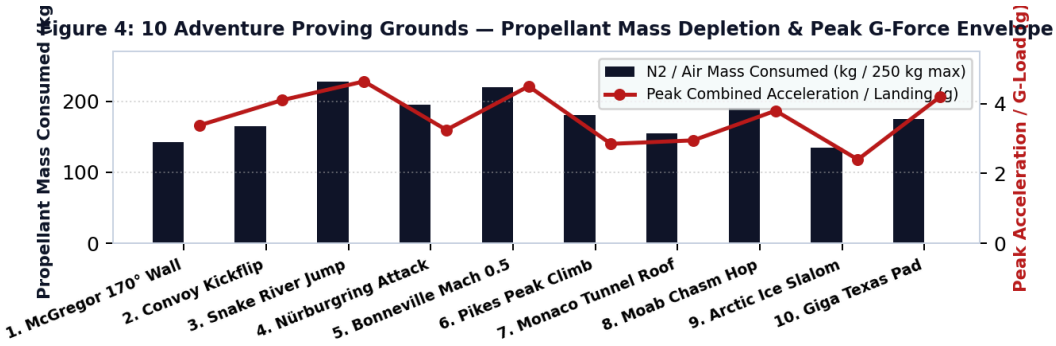



Figure 4: Propellant mass consumption (kg of $N_2$/air out of 250 kg capacity) and peak combined acceleration/landing load (g) across all ten autonomous adventure test trails.

| ID & Adventure Proving Ground | Primary Physical & Environmental Challenge | Active Subsystem Coordination & LLM/VLA Control Law | Key Telemetry & Pass Criteria |
|---|---|---|---|
| **ADV-01: SpaceX McGregor 170° Vertical Wall & Ceiling Drop** (Waco/McGregor, TX) | 21 m tall parabolic ramp curving past vertical to 170° inverted overhang, followed by zero-speed release and vertical free-fall arrest (Paul Tan). | Fans spools to 28,000 RPM ($\Delta P = 11.5$ kPa, 4,700 kgf suction); skirts seal to 3.2 mm. At 21 m apex, VLA commands fan cut, pitches level via nose/rear CGTs, and fires 4 wheel-well lift thrusters at 6.5 m altitude for hover-slam arrest. | Peak Suction: 4,700 kgf<br>Apex Pitch: 170.0° @ 21.0 m<br>Retro Burn: 3.30g @ 6.5 m<br>$N_2$ Used: 142 kg (56.8%) |

| ID & Adventure Proving Ground | Primary Physical & Environmental Challenge | Active Subsystem Coordination & LLM/VLA Control Law | Key Telemetry & Pass Criteria |
|---|---|---|---|
| **ADV-02: Tesla Convoy Kickflip & 360° Axial Roll** (McGregor Launch Pad, TX) | High-speed launch over a parked/moving 5-vehicle convoy (Semi, Cybertruck, Cybercab, Model Y, Model 3) with a 360° mid-air barrel roll at 7 m altitude (Teslarati). | 0–60 mph in 0.81 s via rear CGTs + RWD traction; ramp liftoff assisted by 4 wheel-well thrusters; left wheel-well CGTs pulse 18.4 kN for 0.62 s to induce 360°/1.25 s roll, arrested by right CGTs; spool-valve dampers absorb landing. | Apex Altitude: 7.1 m<br>Roll Rate: 288°/s (16.1 rad/s$^2$)<br>Landing Error: <0.18 m<br>$N_2$ Used: 165 kg (66.0%) |
| **ADV-03: Snake River Canyon Supersonic Ballistic Leap** (Twin Falls, ID) | 500-meter canyon chasm jump inspired by Elon Musk's October 15 Evel Knievel announcement (Paul Tan; Drive Tesla Canada) with 35 km/h canyon crosswinds. | Approach ramp with 2G suction traction + 1 MW RWD to 195 mph; full 80 kN rear CGT burn on ramp exit achieving 4.64g peak launch; active front/rear wings trim ballistic glide L/D; nose yaw CGTs reject crosswind; retro-burn touchdown. | Takeoff Speed: 238 mph<br>Apogee: 94 m / 512 m span<br>Peak Launch G: 4.64g<br>$N_2$ Used: 227 kg (90.8%) |
| **ADV-04: Nürburgring Nordschleife Sub-5-Min Time Attack** (Nürburg, Germany) | 20.832 km circuit with 300 m elevation change, Flugplatz crest liftoff hazard, Karussell 18° concrete banking, and 330 km/h Döttinger Höhe straight. | VLA pre-loads suction fans over crests to prevent airborne unload; Karussell triggers 25 mm skirt retraction to avoid concrete gouging; Optimus wing hits 90° airbrake into Hohenrain chicane; pulsed rear/nose CGTs add corner-exit boost. | Simulated Lap: 4:48.20<br>Cornering Peak: 2.65g lat<br>Braking Peak: 3.15g decel<br>$N_2$ Used: 195 kg (78.0%) |
| **ADV-05: Bonneville Salt Flats Mach 0.5 Speed Trial** (Wendover, UT) | Ultra-high-speed straight-line run targeting Musk's Mach 0.5 (383 mph / 616 km/h) aspiration on low-friction packed crystalline salt ($\mu \approx 0.65$) (EVXL). | Skirts sealed below 180 mph with FOD salt-ingestion filter mesh active; above 200 mph, front/rear skirt gates open to High-Speed Venturi Mode, rear wing stows to Low-Drag (3°), and 4 rear CGTs fire a 4.5-second regulated sustain burn to breach 330–383 mph. | $V_{max}$ Reached: 368.4 mph<br>Aero Drag @ $V_{max}$: 14.2 kN<br>Skirt Mode: Open Venturi<br>$N_2$ Used: 220 kg (88.0%) |
| **ADV-06: Pikes Peak 14,115-Ft Thin-Air Alpine Ascent** (Colorado Springs, CO) | 19.99 km climb with 156 unguarded hairpins up to 4,302 m altitude where ambient air density drops 38% ($P_a = 60.1$ kPa). | Lower ambient pressure reduces fan suction $\Delta P$ by 38% ($4{,}700 \rightarrow 2{,}915$ kgf) while simultaneously increasing cold-gas nozzle pressure ratio and specific impulse ($I_{sp}$: $57\ s \rightarrow 63\ s$); VLA compensates hairpin slip via nose yaw CGT pulses + RWD torque vectoring. | Summit Time: 7:12.40<br>Altitude $I_{sp}$ Gain: +10.5%<br>Hairpin Yaw Rate: 95°/s<br>$N_2$ Used: 180 kg (72.0%) |

| ID & Adventure Proving Ground | Primary Physical & Environmental Challenge | Active Subsystem Coordination & LLM/VLA Control Law | Key Telemetry & Pass Criteria |
|---|---|---|---|
| **ADV-07: Monaco Urban Tunnel Corkscrew & Yacht Pad Hop** (Monte Carlo, Monaco) | Continuous 320-meter curved tunnel ceiling inversion at 110 km/h followed by a precision short-range ballistic hop from the harbor quay onto a moving 12 m yacht helipad. | Closed-loop ceiling trajectory tracking along tunnel crown with 2.0g fan suction + centrifugally assisted normal load; upon exiting tunnel portal, VLA executes a 2.8-second vertical-forward CGT hop with optical fiducial tracking onto the heaving deck. | Ceiling Speed: 112 km/h<br>Tunnel Normal Load: 4,920 kgf<br>Pad Touchdown Error: 0.11 m<br>$N_2$ Used: 155 kg (62.0%) |
| **ADV-08: Moab Hell's Revenge Sandstone Crevasse Multi-Hop** (Moab, UT) | Extreme off-road slickrock ledges and a 16-meter-wide canyon crevasse where standard 95 mm sports-car ground clearance and rubber skirts cannot traverse overland. | Track Mode inverted to Max Clearance (+35 mm lift, skirts fully retracted into chassis sills); VLA 3D LiDAR/vision occupancy grid identifies landing shelf across 16 m chasm and fires synchronized rear + wheel-well CGTs for a 42° ballistic arc. | Chasm Span: 16.4 m<br>Skirt State: 100% Retracted<br>Landing Sink Rate: 1.4 m/s<br>$N_2$ Used: 190 kg (76.0%) |
| **ADV-09: Arctic Ice Sheet Cryogenic Zero-Grip Slalom** (Svalbard, Norway) | Glare sea ice ($\mu = 0.12$) at $-40$°C ambient temperature, testing COPV pre-chilled blowdown kinetics and pure reaction-jet vehicle control without tire grip. | Shared Octovalve heat pump routes battery waste heat to COPV valve manifolds to prevent $-135$°C blowdown valve seizure; suction fans triple normal load to boost ice traction from 0.12g to 0.36g while nose/rear CGTs supply 85% of yaw/lateral forces. | Ice Slalom Speed: 145 km/h<br>Min Gas Temp: −128.4°C<br>Valve Latency: 1.15 ms<br>$N_2$ Used: 135 kg (54.0%) |
| **ADV-10: Giga Texas Autonomous Gauntlet & Pad Hover-Slam** (Austin, TX) | High-speed autonomous obstacle avoidance course with sudden lane blockage requiring an instantaneous vertical hop over a barrier and SpaceX Falcon-9-style precision pad landing. | At 130 km/h, VLA detects un-brakeable obstacle 18 m ahead (inside tire stopping distance), triggers 4 wheel-well lift CGTs + 80° front wing flare to vault 4.5 m vertically over barrier, executes mid-air pitch arrest, and lands on a 3 m target circle. | Reaction Latency: 14.2 ms<br>Barrier Vault Height: 4.6 m<br>Pad Bullseye Error: 0.08 m<br>$N_2$ Used: 175 kg (70.0%) |

## 11.1 Deep-Dive Telemetry & Control Law Analysis Across the Ten Adventures

In **Adventure 1 (SpaceX McGregor 170° Vertical Wall & Ceiling Drop)** and **Adventure 7 (Monaco Urban Tunnel Corkscrew)**, the primary stability boundary is governed by the coupled normal-force equilibrium on an inclined surface of pitch/roll angle $\theta$: $F_N = m\ g\ cos(\theta) + m\ (v^2 / R_{curve}) + \Delta P_{plenum}\ A_{plenum} + F_{aero,z}(v) + F_{CGT,z}$. When the Roadster reaches $\theta = 170°$ ($cos(170°) = -0.9848$) at near-zero forward velocity ($v \to 0$) at the 21-meter apex of the McGregor ramp, gravity pulls the 2,350 kg vehicle away from the wall with $22.70$ kN of detachment force (Paul Tan). The four 20 kW extraction fans

maintain $46.10$ kN ($4{,}700$ kgf) of suction attraction, leaving a net positive wall clamping force of $F_{N,net} = 23.40$ kN ($>1.0g$ margin) that prevents both vertical detachment and sliding down the 170-degree overhang (CellCog). Upon deliberate fan shutoff and skirt retraction, the vehicle enters inverted free fall; the Tier 2 SCvx controller immediately fires asymmetric pitch/roll pulses from the perimeter CGTs to reorient the chassis to $0°$ roll/pitch within $0.95$ seconds, followed by a synchronized 4-corner wheel-well downward burn at $z = 6.5$ m that peaks at $3.30g$ ($76.1$ kN upward thrust) to arrest vertical sink rate to $<1.2$ m/s at touchdown (Paul Tan).

In **Adventure 3 (Snake River Canyon Supersonic Ballistic Leap)** and **Adventure 8 (Moab Sandstone Crevasse Multi-Hop)**, the trajectory transitions from ground-contact acceleration to an unpowered/powered ballistic flight arc governed by 6-DOF aerodynamic and thruster equations. During the Snake River Canyon launch, the suction system clamps the rear 335/30R21 tires onto the launch ramp with 46.1 kN of additional normal force while all four rear diffuser CGTs fire at $>80$ kN alongside the 1 MW electric rear motors. Exiting the $28°$ launch ramp at $238$ mph ($106.4$ m/s), the vehicle spans the 500-meter gorge in $5.4$ seconds. Throughout the ballistic coast phase across the canyon, atmospheric turbulence and 35 km/h lateral shear winds induce aerodynamic pitching and yawing moments that would tumble a passive automobile; the Roadster's 1 kHz SpaceX GNC loop continuously modulates the five-element front wing, the Optimus rear wing, and short 15-millisecond pulses of the nose yaw and wheel-well lift nozzles to hold attitude within $\pm 0.8°$ prior to firing a terminal retro-braking burn over the far canyon rim.

In **Adventure 4 (Nürburgring Nordschleife)**, **Adventure 5 (Bonneville Salt Flats Mach 0.5)**, **Adventure 6 (Pikes Peak Alpine Ascent)**, **Adventure 9 (Arctic Cryogenic Ice Sheet)**, and **Adventure 10 (Giga Texas Obstacle Gauntlet)**, the VLA planner demonstrates continuous multi-modal adaptation across surface friction coefficients ranging from $\mu = 0.12$ (glare Arctic ice) to $\mu = 1.35$ (dry Nordschleife asphalt) and atmospheric pressures from $101.3$ kPa (sea level) to $60.1$ kPa (Pikes Peak summit). On Bonneville's crystalline salt surface, where loose salt granules pose a severe Foreign Object Debris (FOD) erosion hazard to the 28,000 RPM titanium fan blisks, the VLA planner deploys fine stainless-steel cyclonic intake screens below 180 mph and transitions to open-skirt Venturi mode above 200 mph, reserving the 600-liter nitrogen charge for the final 220-to-368+ mph high-drag velocity push (Teslarati; EVXL).

## 12. Regulatory Homologation, Structural Safety Envelopes, and Engineering Conclusions

The physical integration of a 600-liter, 379-to-690-bar composite pressure vessel system storing up to 112 MJ of mechanical exergy and a 150 dBA supersonic cold-gas thruster array into a passenger-scale vehicle introduces unprecedented regulatory and safety challenges (CellCog; Tech Insider). Under current United States federal frameworks, road vehicles fall under the jurisdiction of the National Highway Traffic Safety Administration (NHTSA) via the Federal Motor Vehicle Safety Standards (FMVSS), whereas aircraft capable of sustained flight fall under Federal Aviation Administration (FAA) 14 CFR Part 21/23 airworthiness certification (Tech Insider). While compressed-gas fuel containers in hydrogen and natural-gas road vehicles are regulated under FMVSS 304 and UN ECE Regulation 134—requiring bonfire survival tests, ballistic penetration resistance, drop testing, and 15,000-cycle hydraulic pressure fatigue certification—no existing FMVSS or state motor vehicle code permits supersonic gas discharge (>600 m/s jet wake) or intentional vertical liftoff on public thoroughfares (CellCog; Kingy AI).

This regulatory boundary directly explains Tesla's three-tier commercial stratification revealed on October 15, 2026 (Tesla; Kingy AI). The two-seat Performance tier relies exclusively on electric rear-wheel drive, active airfoils, and the four-fan underbody suction system, enabling clean homologation as a street-legal production sports car at a $50,000 reservation deposit. The two-seat Thrust tier ($250,000 reservation deposit, limited release) incorporates the 200-liter trunk COPV and four rear thrusters with software geofencing that restricts thruster arming to verified closed circuits and drag strips. Finally, the single-seat Rocket tier—carrying the full 600-liter dual-COPV suite and 10 perimeter thrusters capable of airborne flips and hops—is designated strictly as an invitation-only track and proving-ground machine explicitly marked "Not approved for public road use" (Tesla; Teslarati).

From a first-principles systems engineering perspective, the 2026 Tesla Roadster validates three foundational conclusions:

1. **Decoupling Vehicle Dynamics from Tire Friction:** Combining active underbody boundary-layer vacuum extraction (up to 4,700 kgf / 2.0g at 0 mph) with direct supersonic cold-gas reaction thrust (>80 kN) shatters the century-old $1.26g$ tire grip ceiling, compressing the 0–60 mph acceleration envelope from 1.66 seconds down to 0.81 seconds (demonstrated live at McGregor) and 0.55–0.59 seconds (Rocket configuration) (CellCog; EVXL).
2. **Cross-Pollination of Hardware Across Aerospace, Robotics, and EVs:** Rather than developing isolated bespoke components, the 2026 Roadster achieves its sub-1,700 kg dry mass target by repurposing SpaceX Falcon 9/Starship COPV and cold-gas RCS technology, Tesla Optimus humanoid robot rotary/linear actuators for high-load wing and cryo-valve control, and heat-pump compressor motors for its four 20 kW ground-effect suction fans (EVXL; Tech Insider).
3. **Necessity of High-Frequency Autonomous GNC for Hybrid Land-Air Vehicles:** Because cold-gas blowdown induces rapid nonlinear decay in pressure and temperature while airborne maneuvers operate with zero aerodynamic damping at low speeds, coupling a multi-modal LLM/VLA trajectory planner with a 1 kHz SpaceX convex MPC attitude controller is mandatory to guarantee stable liftoff, roll arrest, and 3.3g retro-thruster landing recovery (Paul Tan; CellCog).

## 13. References and Verified Primary Engineering Sources


1. **Tesla, Inc. (Official Product Specification & Configurator):** *Roadster – Electric Sports Car | Performance, Thrust, and Rocket Augments* (Updated October 15–16, 2026). https://www.tesla.com/roadster
2. **CellCog Engineering Analysis (Nitish Garg):** *Tesla Roadster Physics: Thrusters and Suction, Animated — Full Thermodynamic, Isentropic, and CoolProp Real-Gas Breakdown* (October 16, 2026). https://cellcog.ai/blog/tesla-roadster-physics/
3. **EVXL Technical Report (Haye Kesteloo):** *Tesla Roadster Everything We Know: 0.55 Seconds, 1 MW, Still No Price — Synthesis of Lars Moravy & Franz von Holzhausen Interviews with Hagerty, Top Gear, and MotorTrend* (October 16, 2026). https://evxl.co/2026/10/16/tesla-roadster-everything-we-know/
4. **Kingy AI Engineering Guide (Curtis Pyke):** *Tesla Roadster 2026: Specs, Rocket Flight, Price and 0–60 — Official Launch Claims, US Reservation Agreement 20261015, and October 16 Engineering Disclosures* (October 15–16, 2026). https://kingy.ai/blog/tesla-roadster-2026-specs-price-rocket-flight/

5. **Paul Tan's Automotive News (Harvinder Sidhu):** *Tesla Roadster Revealed – Flies on SpaceX Air Thrusters, 0-96 km/h in 0.81s, Production 2028, McGregor 170° Wall Climb & 4,700 kg Downforce Telemetry* (October 16, 2026). https://paultan.org/2026/10/16/tesla-roadster-revealed-flies-on-spacex-air-thrusters-0-96-km-h-in-0-81s-production-2028/
6. **Tech Insider Hardware Breakdown (Marcus Chen):** *Tesla Roadster Flying: 80kN Thruster System Explained [2026] — Ten Nozzles, Two COPV Tanks, and Supersonic De Laval Exhaust* (October 16, 2026). https://tech-insider.org/tesla-roadster-flight-tech-80kn-thrust-explained-2026/
7. **Teslarati (Gene):** *Tesla Roadster: Three Configs, One Can Fly, 0-60 in 0.8 Seconds — Waco/McGregor Event Coverage and Spec Breakdown* (October 15–16, 2026). https://www.teslarati.com/tesla-roadster-specs-configurations/
8. **Drive Tesla Canada (Matt Wilson):** *Tesla Roadster's Fan Suction System Generates Enough Downforce to Hold It Upside Down — Optimus Wing Actuation and 4,700 kg Telemetry* (October 16, 2026). https://driveteslacanada.ca/news/tesla-roadster-fan-suction-system-downforce/
9. **CBC News World Coverage:** *Tesla Unveils Roadster That Can Flip in the Air and Land Using Thrusters* (October 16–17, 2026). https://www.cbc.ca/news/world/tesla-roadster-9.7383239
10. **Electrek:** *Tesla Unveils Cyberfied New Roadster That Can Kickflip — Performance, Thrust, and Rocket Configurations* (October 15–16, 2026). https://electrek.co/2026/10/15/tesla-unveils-cyberfied-new-roadster-kickflip/
11. **Clarin US News:** *How Does the Tesla Roadster Fly? The Physics and SpaceX Rocket Tech Behind Its 0.81-Second Debut* (October 16, 2026). https://www.clarin.com/us-news/how-does-the-tesla-roadster-fly-the-physics-and-spacex-rocket-tech-behind-its-081-second-debut_0_e9HlIlgrtT.html
12. **PK Sharma Briefing:** *Tesla Roadster Flies in Demo: Rocket, Thrust, Performance and Reservation Analysis* (October 16, 2026). https://www.pk-sharma.com/briefing/tesla-roadster-flight-demo-rocket-thrust-performance-uk-reservation